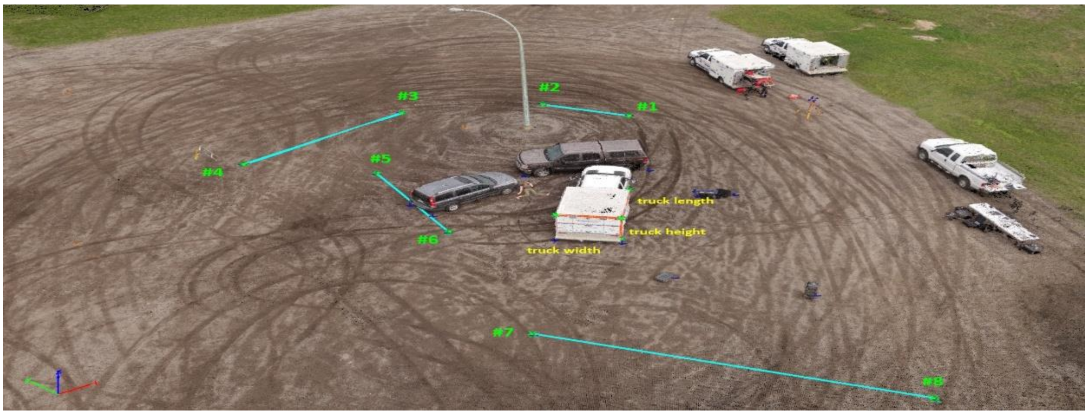
UNIT -01 What is Drone Forensics?



Drone Forensics is a specialized branch of **digital forensics** that focuses on the investigation and analysis of **Unmanned Aerial Vehicles (UAVs)**, commonly known as drones, involved in incidents, accidents, or criminal activities.

It involves the **systematic identification, collection, preservation, examination, analysis, and presentation of both physical and digital evidence** obtained from drones, their controllers, storage media, and associated software.

Key Aspects of Drone Forensics

- Identification of drone **model, make, and manufacturer**
- Extraction and analysis of **flight logs and GPS data**
- Recovery of **images, videos, and telemetry**
- Tracing the **pilot or operator**
- Determining **intent** (legal, illegal, accidental, or malicious)

Why Drone Forensics is Important

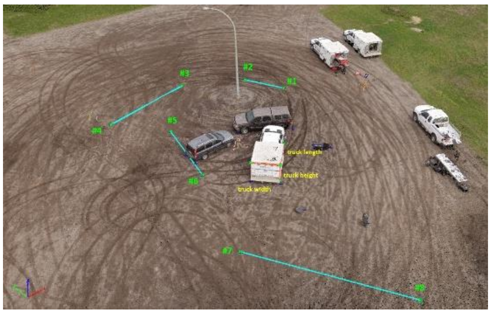
Drone forensics plays a vital role in:

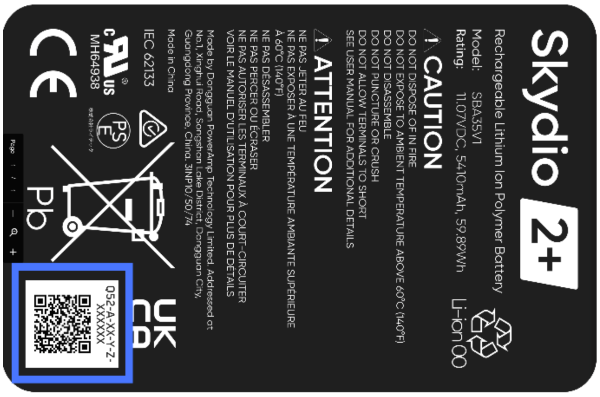
- Terrorism and border security investigations
- Smuggling and illegal payload delivery
- Privacy invasion and surveillance cases
- Airspace violations near airports and sensitive areas
- Accident and crash investigations

Example - If a drone crashes near a restricted military area, drone forensics helps investigators:

- Identify the drone brand and origin
- Recover stored flight data
- Reconstruct the flight path
- Link the drone to its operator
- Present evidence in court

Physical Evidence in Drone Forensics





Physical evidence in drone forensics refers to all **tangible components** of a drone and its accessories that can be collected and examined during an investigation. This evidence plays a crucial role in identifying the drone, reconstructing incidents, and establishing links between the drone, the operator, and the crime.

1. Types of Physical Evidence

1.1 Drone Airframe (Body)

- May be **crashed, damaged, or intact**
- Contains manufacturer labels and serial numbers
- Damage patterns help in **crash and impact analysis**

1.2 Motors and Propellers

- Indicate **direction, speed, and duration of flight**
- Broken or bent propellers help identify **collision events**
- Motor serial numbers may link to a specific model

1.3 Battery (Usually Li-Po)

- Stores **charge cycles and usage data**
- Serial numbers assist in **ownership tracing**
- Damage to battery may indicate **mid-air failure or forced landing**

1.4 Storage Media (SD Card / Internal Memory)

- Contains **raw images, videos, and flight logs**
- Metadata provides **date, time, GPS coordinates**
- Often holds **unaltered primary evidence**

1.5 Camera and Gimbal

- Primary source of **visual surveillance evidence**
- Gimbal orientation data helps reconstruct **camera angle and target**
- Damage may indicate **intentional destruction**

1.6 Remote Controller

- Stores **pairing information and control history**
- Can be linked to a specific drone and operator
- Physical wear may indicate frequent usage

1.7 SIM Card (Advanced Drones)

- Used in **LTE/5G-enabled drones**
- Helps trace **network activity and location**
- Can link drone operations to a **subscriber identity**

1.8 Payloads

- May include **contraband, explosives, drugs, or surveillance tools**
- Strong indicator of **criminal intent**
- Requires careful handling due to **safety risks**

2. Forensic Importance of Physical Evidence

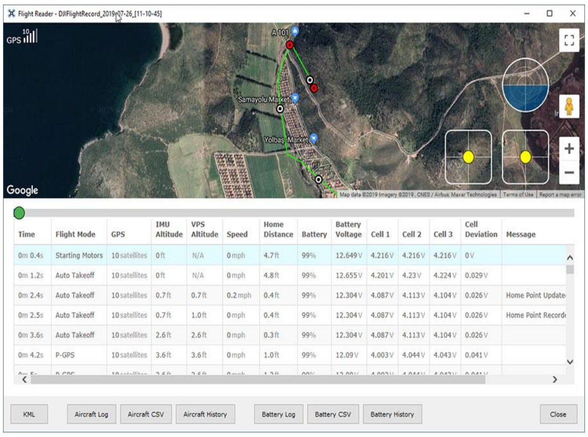
- **Serial numbers** identify manufacturer and supply chain
- **Impact marks** assist in reconstructing crash dynamics
- **Battery data** estimates flight time and distance
- **Storage devices** provide court-admissible digital evidence

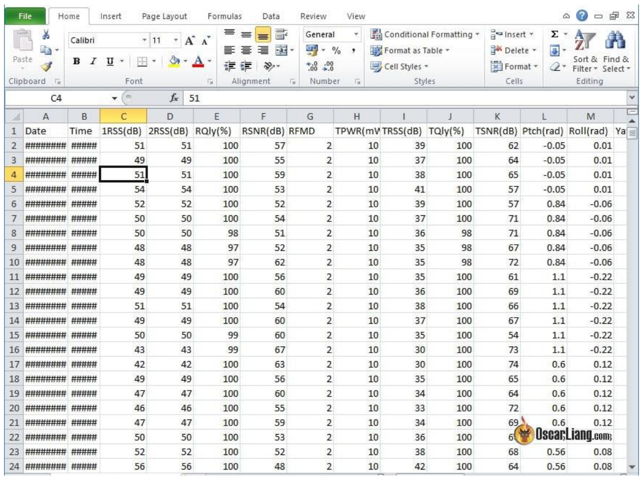
Real-World Example:
In border-smuggling cases, recovered drone batteries and payload attachments helped investigators confirm **long-distance illegal delivery missions**.

3. Handling Physical Evidence

- Photograph evidence **before touching or moving**
- Use **anti-static and Faraday bags**
- Avoid **powering ON** the drone without authorization
- Maintain **chain of custody documentation**
- Store batteries in **fire-safe containers**

DIGITAL TREASURE: UNVEILING DIGITAL EVIDENCE IN DRONE FORENSICS



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4	########	#####	51	51	100	59	2	10	38	100	65	-0.05	0.01	
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In drone forensics, **digital evidence is the true "digital treasure"** hidden within unmanned aerial vehicles (UAVs) and their associated systems. Unlike physical components, digital artifacts reveal **who operated the drone, where it flew, when it was active, and why it was used**. Unveiling this evidence is central to reconstructing drone-related incidents and proving intent in legal investigations.

What is Digital Evidence in Drone Forensics?

Digital evidence refers to any data generated, stored, transmitted, or processed by a drone and its ecosystem, including:

- Onboard memory
- Flight controllers
- Mobile devices
- Cloud platforms
- Ground control systems

This evidence is often **time-stamped, location-aware, and user-linked**, making it highly valuable and legally significant.

Major Sources of Digital Evidence

1. Flight Logs and Telemetry Data

- GPS coordinates, altitude, speed, and direction
- Takeoff and landing locations
- Battery status and flight duration

Forensic Value:- Used to **reconstruct the complete flight path** and prove airspace violations or border intrusions.

Real-World Example:- In restricted airspace violations near airports, flight logs have been used to prove **intentional drone penetration**, leading to criminal prosecution.

2. Images and Video Files

- Photos and videos captured during flight
- Embedded metadata (EXIF data)
- Raw, unedited surveillance footage

Forensic Value:- Provides **visual proof of surveillance, reconnaissance, or payload delivery locations**.

Real-World Example:- Recovered drone footage in smuggling cases revealed **drop zones**, helping authorities arrest ground-level accomplices.

3. Onboard Storage and Memory

- Internal flash memory
- SD cards

Forensic Value:- May contain **deleted files**, cached logs, and historical mission data recoverable using forensic tools.

4. Mobile Applications

Examples include operator control apps used on smartphones or tablets.

- User login credentials
- Flight history
- Cached telemetry data
- Cloud synchronization records

Forensic Value:- Helps **link the drone directly to a specific operator or user account**.

5. Remote Controller and Ground Control Software (GCS)

- Pairing information
- Command logs
- Mission planning data

Forensic Value:- Correlates **operator actions with drone behavior**, strengthening attribution.

6 Cloud and Network Data

- Cloud backups of flight logs
- Account activity records
- Network communication metadata

Forensic Value:
Even if local data is deleted, **cloud artifacts can preserve historical evidence**.

Forensic Analysis of Digital Treasure

Digital evidence analysis involves:

- Secure acquisition using forensic tools
- Hash verification to ensure integrity
- Timeline creation and correlation
- Cross-validation with physical evidence
- Correlation of **telemetry data, mobile logs, and cloud records** creates a legally robust reconstruction of events.

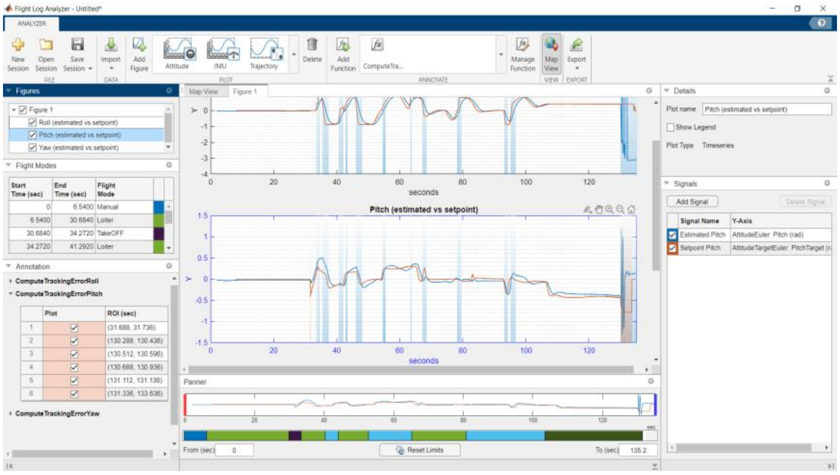
Challenges in Unveiling Digital Evidence

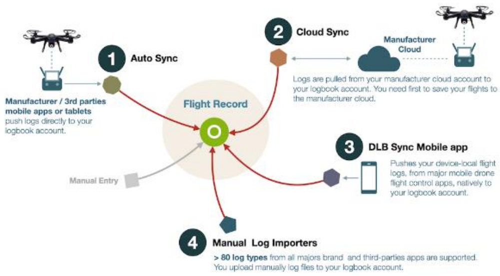
- Encrypted firmware and storage
- Proprietary file formats
- Cloud data jurisdiction issues
- Risk of data overwrite if powered ON improperly

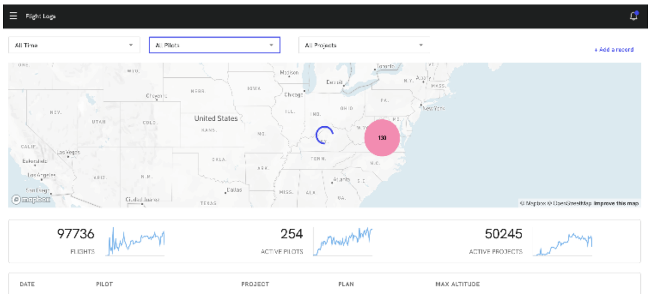
Legal Importance of Digital Evidence

- Establishes **mens rea (criminal intent)**
- Links drone activity to a **specific individual**
- Provides **court-admissible, scientific proof**

Data Storage and Retrieving Methods in Drone Forensics







In drone forensics, understanding **how and where data is stored** and the **methods used to retrieve it** is critical for reconstructing drone activity and establishing legally admissible evidence. Drones generate large volumes of **digital artifacts** across onboard hardware, controllers, mobile devices, and cloud platforms.

Data Storage Locations in Drones

Onboard Internal Memory

- Embedded flash memory inside the flight controller or camera module
- Stores **system logs, configuration files, error logs**, and sometimes media
- Often proprietary and manufacturer-specific

Forensic Importance:
Contains **low-level flight and system data** even when SD cards are removed.

Removable Storage (SD / microSD Card)

- Used for storing **images, videos, and sometimes flight logs**
- Files often include **EXIF metadata** (date, time, GPS coordinates)

Forensic Importance:
Most accessible and frequently holds **raw, unedited evidence**.

Flight Controller Memory

- Stores **telemetry data**, such as altitude, speed, orientation, and GPS
- May retain **historical flight records**

Forensic Importance:
Essential for **flight path reconstruction** and crash analysis.

Remote Controller Storage

- Saves **pairing data, control logs, and timestamps**
- Can be linked to a specific drone and operator

Forensic Importance:
Helps establish **operator–drone association**.

Mobile Device Storage

- Drone control apps store:
    - Flight history
    - Cached logs
    - User credentials
- Data may reside in **app directories, databases, or cache files**

Forensic Importance:
Key source for **operator identification**.

Cloud Storage

- Automatic syncing of:
    - Flight logs
    - Media files
    - User account activity

Forensic Importance:
Preserves evidence even after **local deletion or drone destruction**.

Data Types Stored in Drone Systems

- Flight logs (DAT, TXT, CSV formats)
- GPS and telemetry data
- Images and videos
- Sensor readings
- Error and system logs
- Configuration and firmware files

Data Retrieval Methods in Drone Forensics

1. Manual Extraction

- Removing SD cards and copying data using **write blockers**
- Suitable for **non-damaged drones**

Advantages: Simple and fast
Limitations: Risk of metadata alteration if not handled properly

2. Logical Acquisition

- Accessing data via:
    - USB interfaces
    - Mobile applications
    - Manufacturer tools

Advantages: Non-invasive
Limitations: May not retrieve deleted or hidden data

3. Physical Acquisition

- Bit-by-bit imaging of:
    - Internal memory
    - Flight controller chips

Advantages: Recovers deleted and hidden data
Limitations: Requires expertise and specialized tools

4. Chip-Off Forensics

- Physically removing memory chips from damaged drones
- Used when devices are **burnt, crashed, or water-damaged**

Advantages: Last-resort recovery
Limitations: High risk, irreversible process

5. Mobile Device Forensics

- Extracting drone app data from smartphones
- Includes:
    - App databases
    - Logs
    - Cloud sync records

Advantages: Strong operator attribution
Limitations: Encryption and OS security

6. Cloud Data Retrieval

- Accessed through:
  - Legal requests
  - Account credentials
- Includes historical flight and media data

Advantages: Long-term data retention
Limitations: Jurisdiction and legal constraints

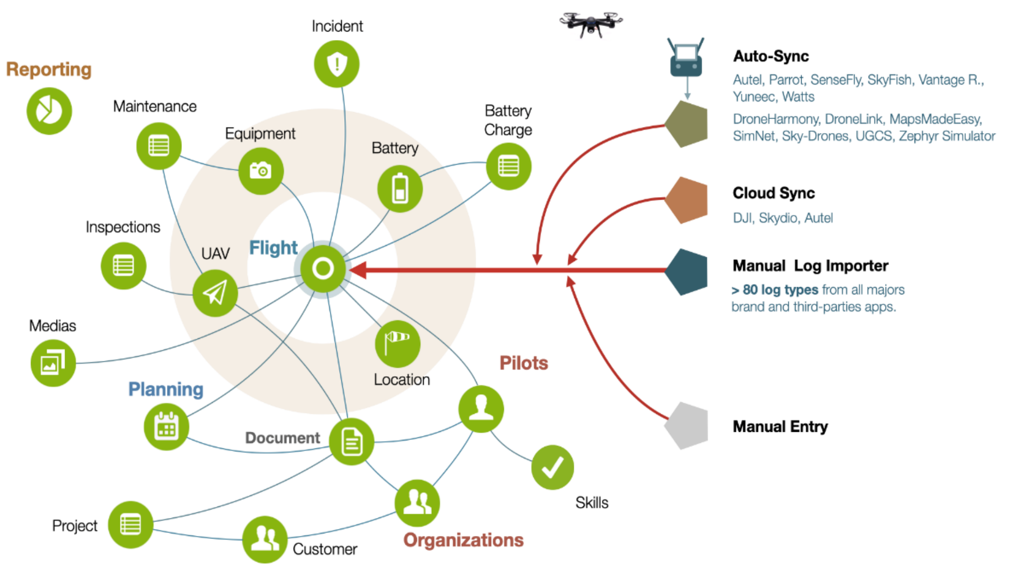
Best Practices for Data Retrieval

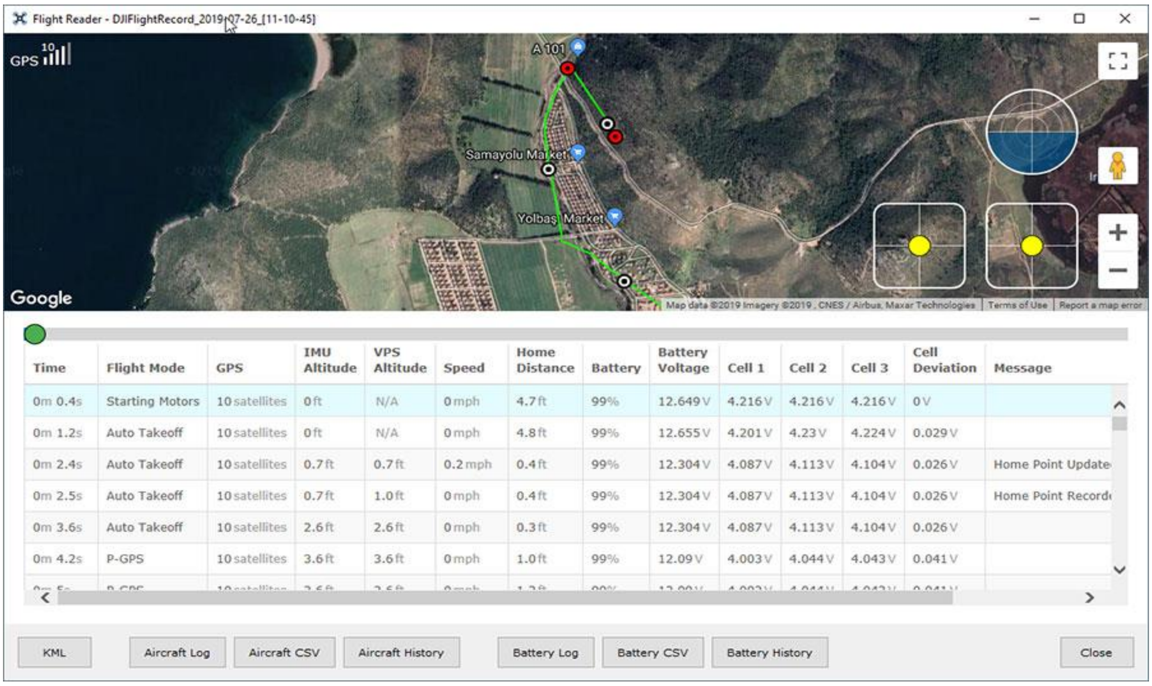
- Never power ON the drone at the scene
- Use forensic write blockers
- Create **hash values** to maintain integrity
- Document every step (chain of custody)
- Correlate data from **multiple sources**

Challenges in Drone Data Retrieval

- Encrypted storage and firmware
- Proprietary file formats
- Automatic data overwrite
- Partial data loss due to crashes

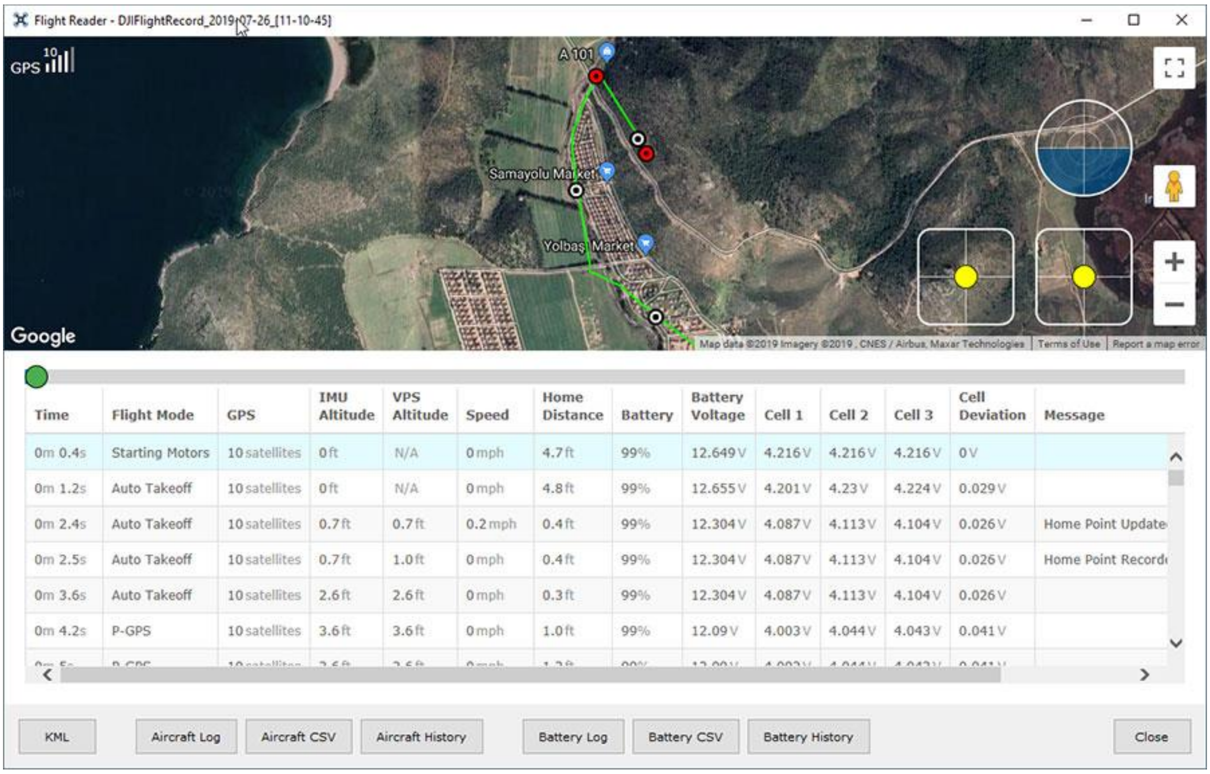
Data storage and retrieval in drone forensics refers to the identification of all locations where drone-related data is stored and the application of forensic acquisition techniques to securely extract, preserve, and analyze that data for investigative and legal purposes.





Drone Incident / Recovery
↓
Scene Documentation
(Photograph, video, location marking)
↓
Evidence Identification
(Drone, controller, SD card, battery, mobile device)
↓
Evidence Preservation
(Anti-static bags, Faraday bags, chain of custody)
↓
Power State Decision
(Do NOT power ON unless authorized)
↓
Data Source Identification
(Onboard memory, SD card, controller, mobile app, cloud)
↓
Data Acquisition Method Selection
(Manual / Logical / Physical / Chip-off)
↓
Forensic Data Extraction
(Write blockers, imaging tools)
↓
Hash Value Generation
(MD5/SHA for integrity verification)
↓
Data Analysis
(Flight logs, GPS, media, telemetry)
↓
Correlation & Reconstruction
(Flight path, timeline, operator link)
↓
Documentation & Reporting
(Forensic report, legal presentation)

Case Study: Drone Investigation Using Recovered Flight Logs



Case Title- Investigation of Unauthorized Drone Activity Using Flight Log Analysis

A quadcopter drone was recovered after a crash near a **restricted border zone**. Authorities suspected the drone was used for **illegal surveillance and smuggling operations**.

Evidence Recovered

- Damaged drone airframe
- Li-Po battery
- microSD card
- Remote controller (not recovered)

Forensic Acquisition Process

1. **Drone powered OFF** and sealed in an anti-static bag
2. SD card removed and imaged using a **write blocker**
3. Internal memory accessed through **logical acquisition**
4. Hash values generated to ensure integrity

Recovered Digital Evidence

- Flight log files containing:
    - GPS coordinates
    - Altitude and speed
    - Time-stamped telemetry
- Battery status logs
- Partial video footage

Flight Log Analysis Findings

- Takeoff point located **5 km from the border**
- Drone crossed restricted airspace at **120 meters altitude**
- Hovering observed near sensitive installations
- Return flight interrupted due to crash

Reconstruction Outcome

- Complete **flight path reconstructed** using GPS telemetry
- Timeline confirmed **intentional border crossing**
- Battery logs indicated **pre-planned long-distance flight**
- Camera angle data suggested **surveillance activity**

Legal Significance

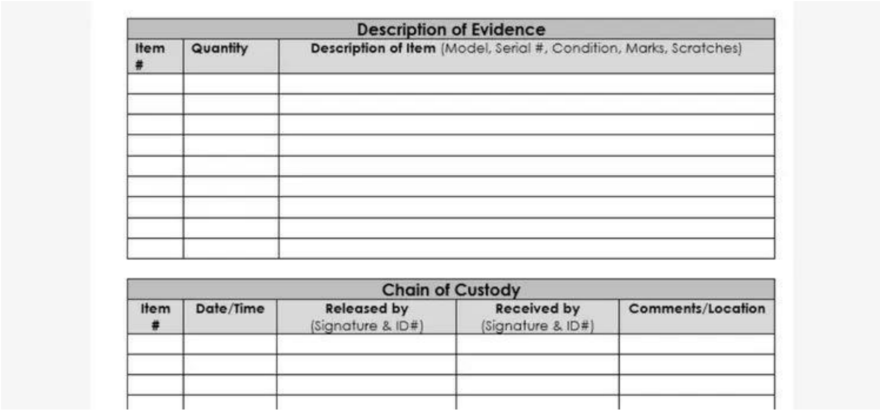
- Flight logs proved **mens rea (criminal intent)**
- GPS data established **violation of restricted airspace**
- Logs became **primary digital evidence** in prosecution

Even without the operator or controller, **flight logs alone were sufficient** to:

- Reconstruct events
- Prove intent
- Support legal action

This demonstrates why **flight logs are considered the “black box” of drones**.

Securing the Secret: Preserving Evidence in Drone Forensics



Description of Evidence		
Item #	Quantity	**Description of Item** (Model, Serial #, Condition, Marks, Scratches)

Chain of Custody				
Item #	Date/Time	Released by (Signature & ID#)	Received by (Signature & ID#)	Comments/Location

Drone forensics involves highly sensitive **physical and digital evidence**. Securing this “secret” evidence is essential to prevent **tampering, loss, or legal rejection**. Proper preservation ensures that drone evidence remains **authentic, intact, and court-admissible**.

Maintenance of Drone Evidence (Physical Evidence Handling)

1. What is Maintenance of Drone Evidence?

Maintenance of drone evidence refers to the **systematic care, storage, documentation, and protection** of all physical components of a drone from the time of seizure until final legal disposal.

2. Steps in Maintaining Drone Physical Evidence

- **Scene documentation** (photographs, videos, GPS location)
- Labeling each component (drone body, battery, SD card, controller)
- Using **anti-static bags** for electronic parts
- Using **fire-proof containers** for Li-Po batteries
- Storing evidence in **secure, access-controlled lockers**

3. Importance

- Prevents physical damage and contamination
- Preserves serial numbers and identifiers
- Supports accurate crash and damage reconstruction

Real-World Insight- Improper battery storage has led to evidence destruction due to fire**, resulting in loss of critical forensic data.**

Chain of Custody (CoC) in Drone Forensics

1. What is Chain of Custody?

The **Chain of Custody (CoC)** is a **legal document and process** that records every individual who handled drone evidence, from seizure to courtroom presentation.

2. Chain of Custody in Drone FS

Each transfer of drone evidence must record:

- Date and time
- Name and signature of handler
- Purpose of transfer
- Condition of evidence

3. Importance of CoC

- Ensures **evidence integrity**
- Prevents allegations of tampering
- Makes evidence **legally admissible**

Example- If a drone SD card is analyzed without CoC documentation, the defense can claim data manipulation**, leading to rejection of evidence.**

Preserving Digital Evidence in Drone Forensics

1. What is Digital Evidence Preservation?

Digital evidence preservation ensures that **flight logs, GPS data, images, videos, and app data** remain unchanged from the moment of acquisition.

2. Methods of Preserving Digital Evidence

- Never power ON the drone at the scene
- Use **write blockers** during data extraction
- Create **forensic images (bit-by-bit copies)**

- Generate **hash values (MD5/SHA)**
- Store original data in **read-only format**

3. Importance

- Prevents data overwrite
- Maintains originality
- Protects against accidental modification

Flight logs are volatile; powering ON a drone can **overwrite last-flight data**, destroying critical evidence.

Maintenance of Chain of Custody (CoC Maintenance)

1. What is CoC Maintenance?

CoC maintenance refers to **continuous updating, verification, and safeguarding** of custody records throughout the investigation.

2. Best Practices for CoC Maintenance

- Use **unique evidence IDs**
- Maintain both **physical and digital logs**
- Restrict access to authorized personnel only
- Record every analysis step
- Secure CoC records against alteration

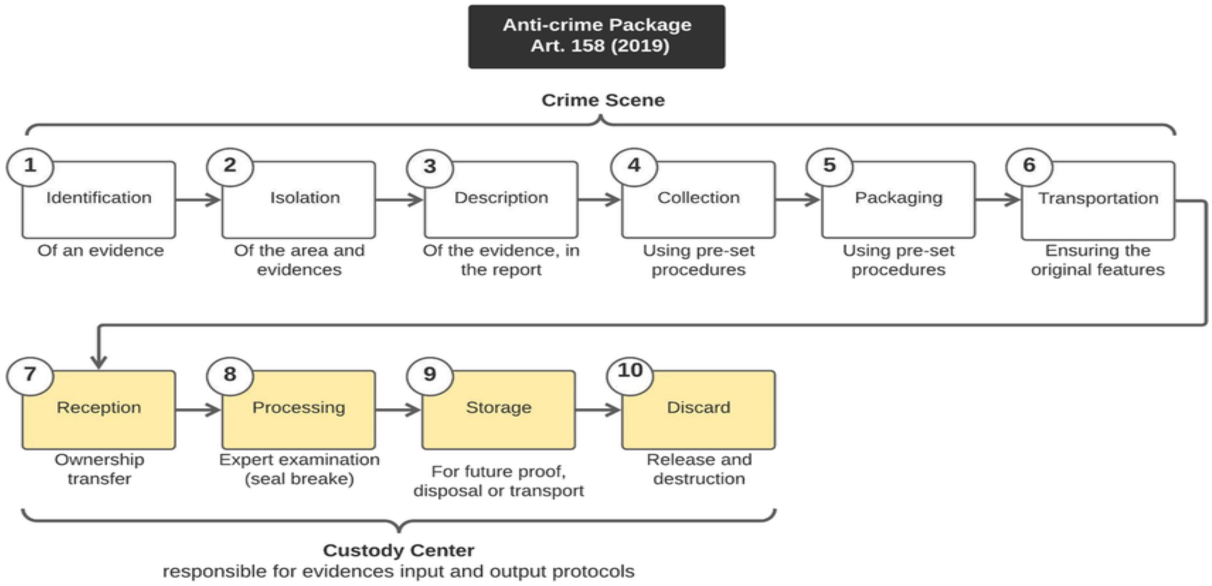
3. Consequences of Poor CoC Maintenance

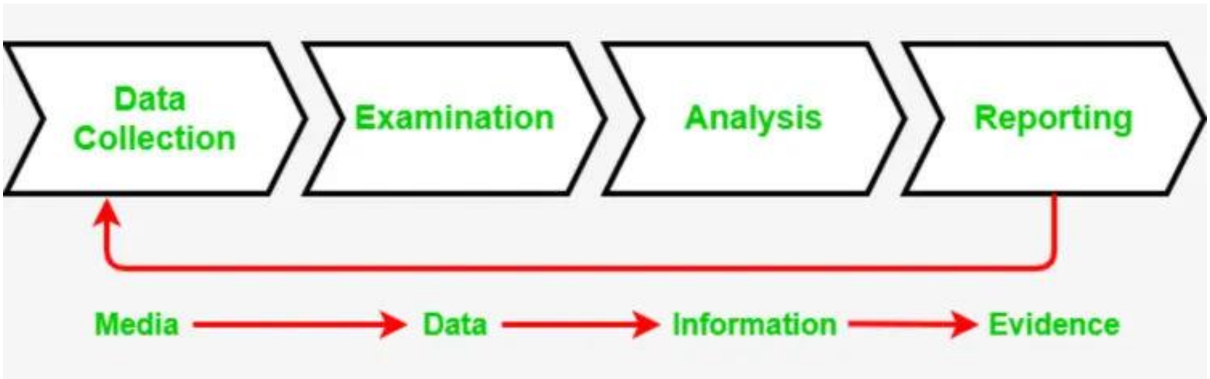
- Evidence rejection in court
- Case dismissal
- Loss of investigator credibility

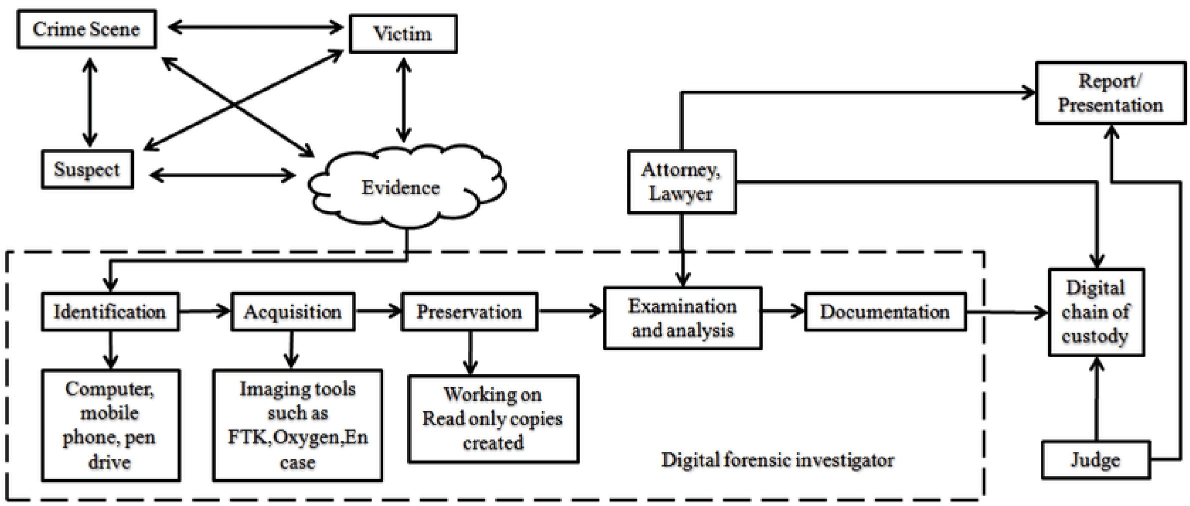
Securing the Secret – Integrated View

Aspect	Purpose
Physical Evidence Maintenance	Prevent damage and contamination
Digital Evidence Preservation	Prevent data alteration
Chain of Custody	Legal accountability
CoC Maintenance	Continuous evidence validation

Flowchart:


Anti-crime Package
Art. 158 (2019)
Crime Scene
1 Identification
Of an evidence
2 Isolation
Of the area and evidences
3 Description
Of the evidence, in the report
4 Collection
Using pre-set procedures
5 Packaging
Using pre-set procedures
6 Transportation
Ensuring the original features
7 Reception
Ownership transfer
8 Processing
Expert examination (seal breake)
9 Storage
For future proof, disposal or transport
10 Discard
Release and destruction
Custody Center
responsible for evidences input and output protocols


Data Collection
Examination
Analysis
Reporting
Media
Data
Information
Evidence


Crime Scene
Victim
Suspect
Evidence
Report/ Presentation
Attorney, Lawyer
Identification
Acquisition
Preservation
Examination and analysis
Documentation
Digital chain of custody
Computer, mobile phone, pen drive
Imaging tools such as FTK,Oxygen,En case
Working on Read only copies created
Digital forensic investigator
Judge

Flowchart

Drone Incident / Recovery
↓
Scene Securing
(Restrict access, ensure safety)
↓
Initial Documentation
(Photographs, videos, GPS location, notes)
↓
Evidence Identification
(Drone, battery, SD card, controller, mobile device)
↓
Evidence Labeling
(Unique ID, date, time, case number)
↓
Evidence Packaging
(Anti-static bags, Faraday bags, fire-safe battery storage)
↓
Chain of Custody Form Initiation
(Collector name, signature, condition)
↓
Evidence Transport
(Secure and sealed transfer)
↓
Evidence Storage
(Access-controlled evidence locker)
↓
Forensic Examination
(Logged access, purpose recorded)
↓
Digital Evidence Imaging
(Write blockers, hash generation)
↓
CoC Update After Each Handling
(Date, time, handler, reason)
↓
Analysis & Reporting
(Findings documented, integrity verified)
↓
Court Presentation
(CoC records submitted with evidence)
↓
Evidence Retention / Disposal
(As per legal guidelines)